# Replication of Roberts et al.'s "Structural Topic Models for Open-Ended Survey Responses"

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# Introduction

- Problem: Open-ended survey responses are rarely analyzed in political science because they are expensive to code and usually rely on human coders
- Roberts et al. (2014) argue that STM offers a semiautomated alternative that can incorporate covariates (party, gender, treatment assignment) and can create topics and estimate treatment effects.
- Paper tries to show that open-ended responses can be analyzed systematically, which makes text analysis more efficient.

# Methods

- Authors used multiple datasets to test STM:
  - Gadarian immigration experiment
  - Rand public goods experiments
  - ANES most important problem data

- In using STM, the authors first used semantic coherence and exclusivity plots to determine best models and then inspected the content of the texts to check that the estimated topics were fitting. After multiple tries with candidate models, they chose the one that best fit.

# Gadarian Experiments: Figure 4: Vocab in Figures 1 and 2

**FIGURE 4 Vocabulary Associated with Topics 1 and 2**

Topic 1:
illeg, job, immigr, tax, pai, american,
care, welfar, crime, system, secur,
social, cost, health, servic, school,
languag, take, us, free

Topic 2:
immigr, illeg, legal, border, need,
worri, mexico, think, countri, law,
mexican, make, america, worker, those,
american, fine, concern, long, fenc

Original

Topic 2
immigr, illeg, legal, think, system, cost, school, problem
hospit, mean, veri, caus, good, illeg, even, rise
caus, good, afford, ancestor, bad, burden, california, citizenship
illeg, immigr, legal, think, hospit, caus, concern, worker

Topic 1
job, tax, american, pai, take, english, worri, secur
us, awai, languag, assimil, drug, fact, tax, servic
amongst, attitud, base, bit, card, citi, custom, decreas
job, servic, tax, worri, crime, pai, american, assimil

Replication

# Gadarian Experiment: Figure 5: Representative Response From Topic 1

**FIGURE 5 A Representative Response from Topic 1**

problems caused by the influx of illegal immigrants who are crowding our schools and hospitals, lowering the level of education and the quality of care in hospitals.

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crime lost jobs benefits paid to illegals health care and food....we cannot feed the world when we have americans starving, etc.

Original

loss of jobs for americans increase in crime challenge on services such as healthcare

---

taking our jobs, not paying taxes, using public services for free

Replication

# Gadarian Experiment: Figure 6: Representative Response from Topic 2

**FIGURE 6 A Representative Response from Topic 2**

i worry about the republican party doing something very stupid. this country was built on immigration, to deny anyone access to citizenship is unconstitutional. what happened to give me your poor, sick, and tired?

---

border control, certain illegal immigrants tolerated, and others immediately deported.

Original

the amount of illegal immigration, and exceptions to the legal immigration quotas that happen all the time. i'm concerned that abuses will sink us soon.

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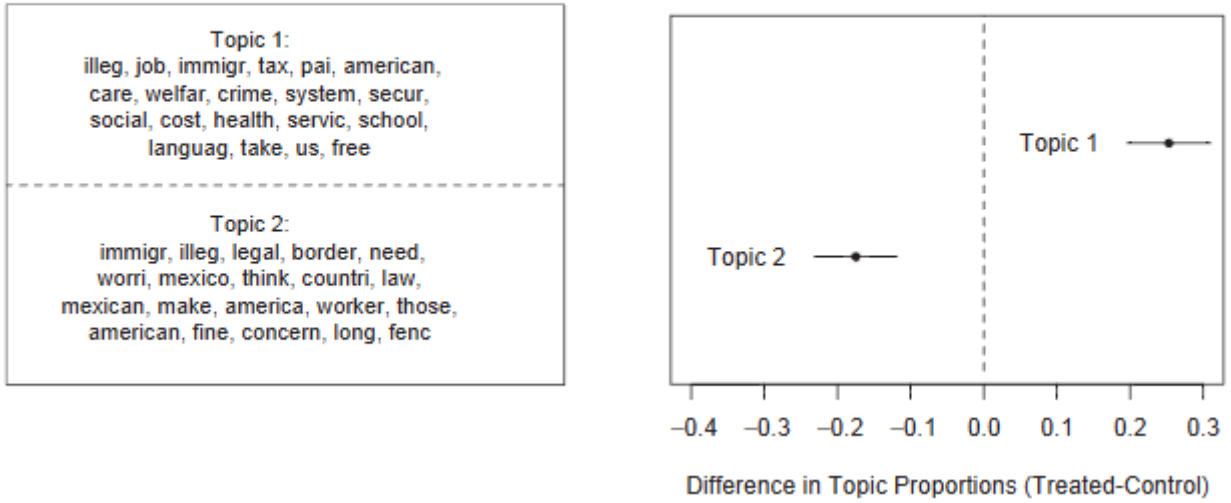
mexico, illegal workers, difficult immigration process

Replication

# Results: Gadarian Experiments: Figure 7

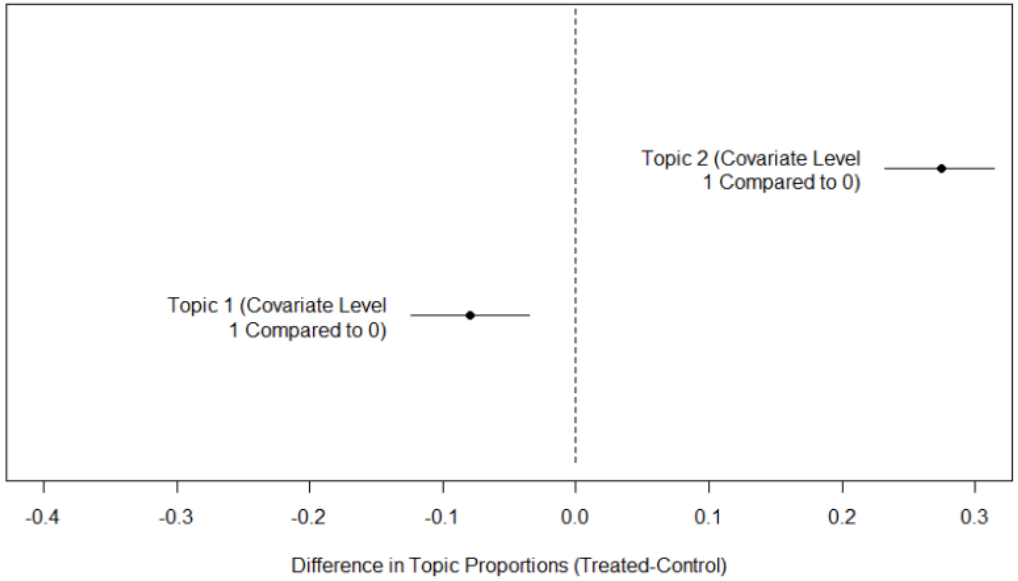
Original

**FIGURE 7 Words and Treatment Effect Associated with Topic 1**



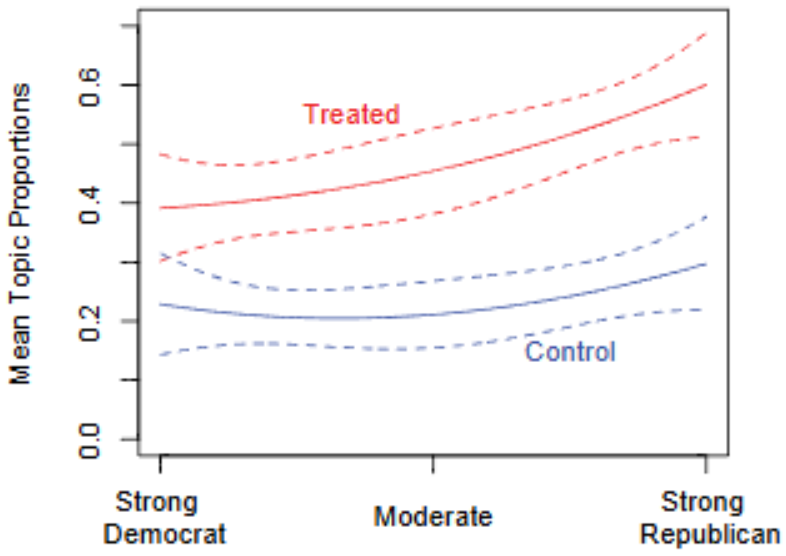

Replication

```
Topic 1 Top Words:
        Highest Prob: immigr, illeg, legal, think, system, cost, school
        FREX: hospit, mean, veri, caus, good, illeg, even
        Lift: caus, good, afford, ancestor, bad, burden, california
        Score: illeg, immigr, legal, think, hospit, caus, concern
Topic 2 Top Words:
        Highest Prob: job, tax, american, pai, take, english, worri
        FREX: us, awai, languag, assimil, drug, fact, tax
        Lift: amongst, attitud, base, bit, card, citi, custom
        Score: job, servic, tax, worri, crime, pai, american
```

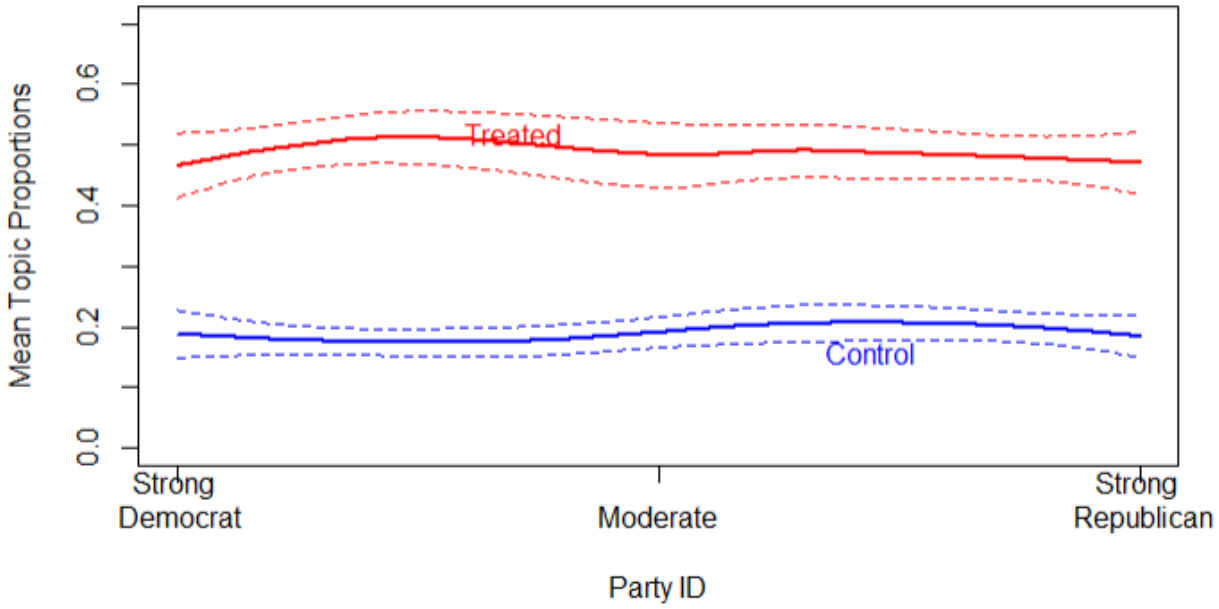


# Gadarian Experiment: Figures 8

**FIGURE 8 Party Identification, Treatment, and the Predicted Proportion in Topic 1**




Original




Replication

# Gadarian Experiment: Figures 9 and 10

**FIGURE 9 Fearful Response with High Topic 1**

"welfare, schools, medical expenses, jobs, crime, housing, driving no insurance or license.,"

**FIGURE 10 Fearful Response with Low Topic 1**

"border control, certain illegal immigrants tolerated, and others immediately deported."

Original

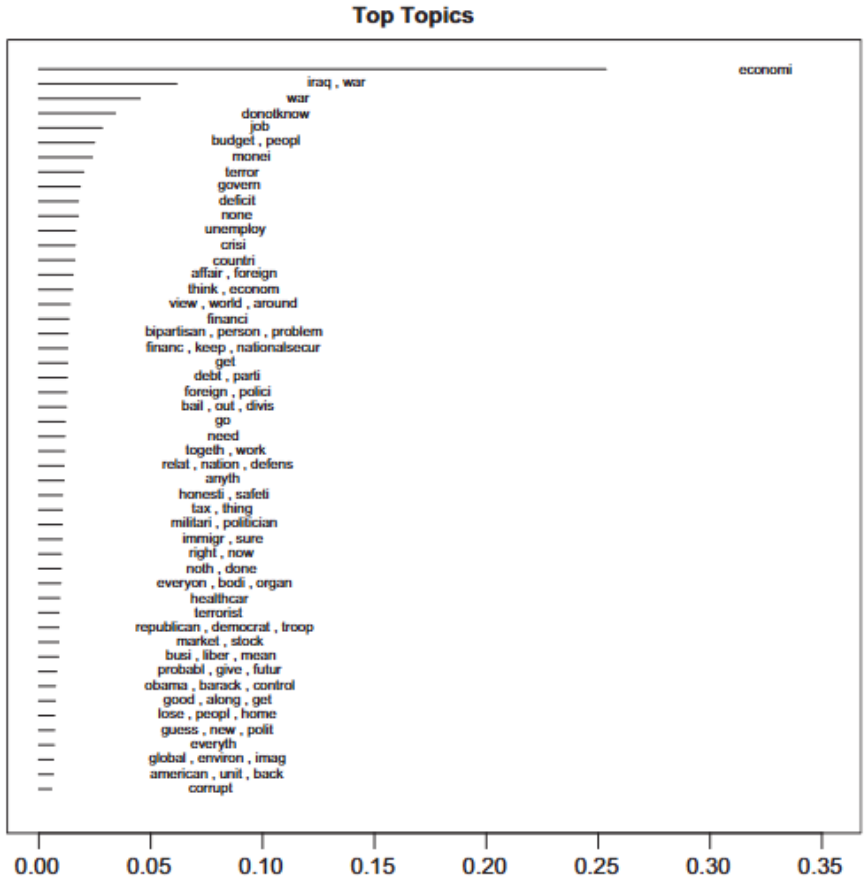
loss
of jobs
for
americans
increase
in
crime
challenge
on
services
such as
healthcare

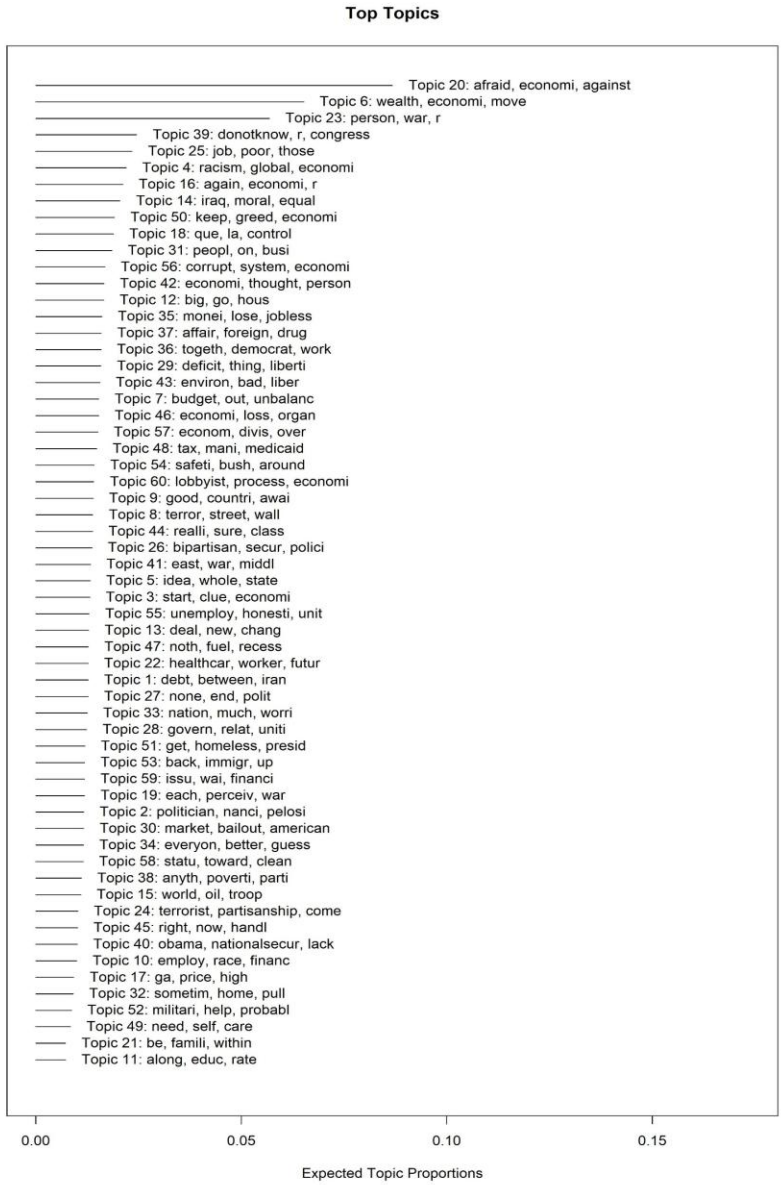
wide open borders that allow
terrorists to enter at will
increased costs of health care due
to illegal immigrants taxing the
hospital system talk of amnesty
causing a greater influx of
illegal immigrants on the other
hand, tough immigration practices
causing families to be torn apart
when illegals are sent home prices
rising dramatically due to the loss
of inexpensive labor

Replication

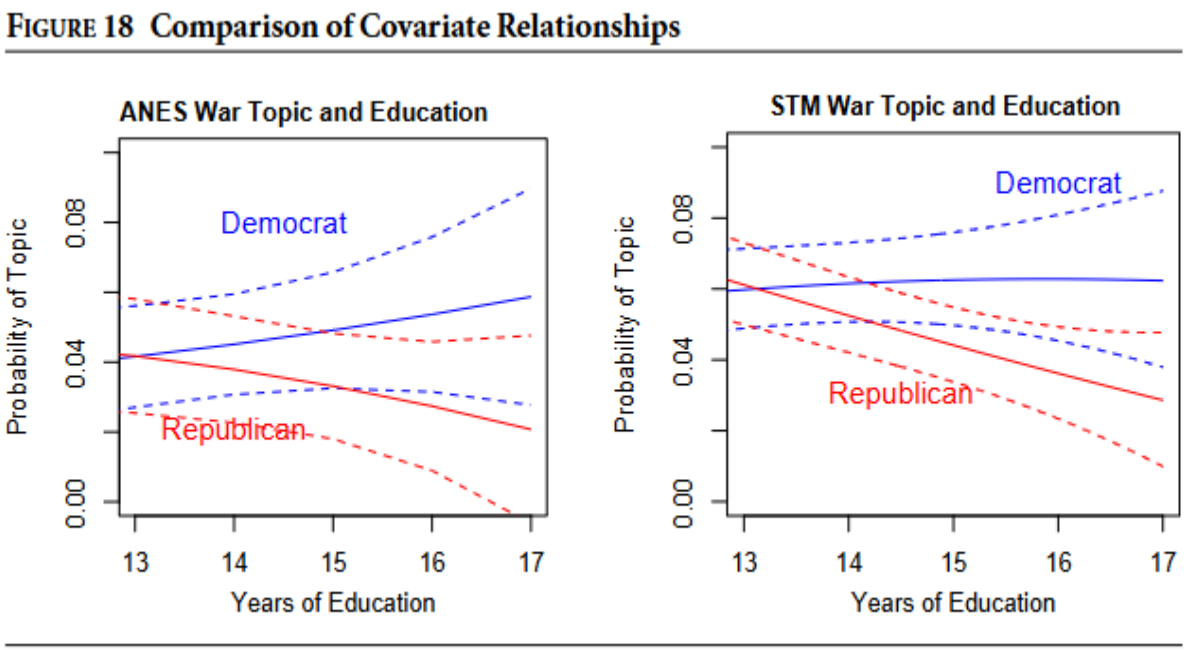
# Results: ANES Most Important Problem: Figure 17

**FIGURE 17 STM Topics from ANES Most Important Problem**

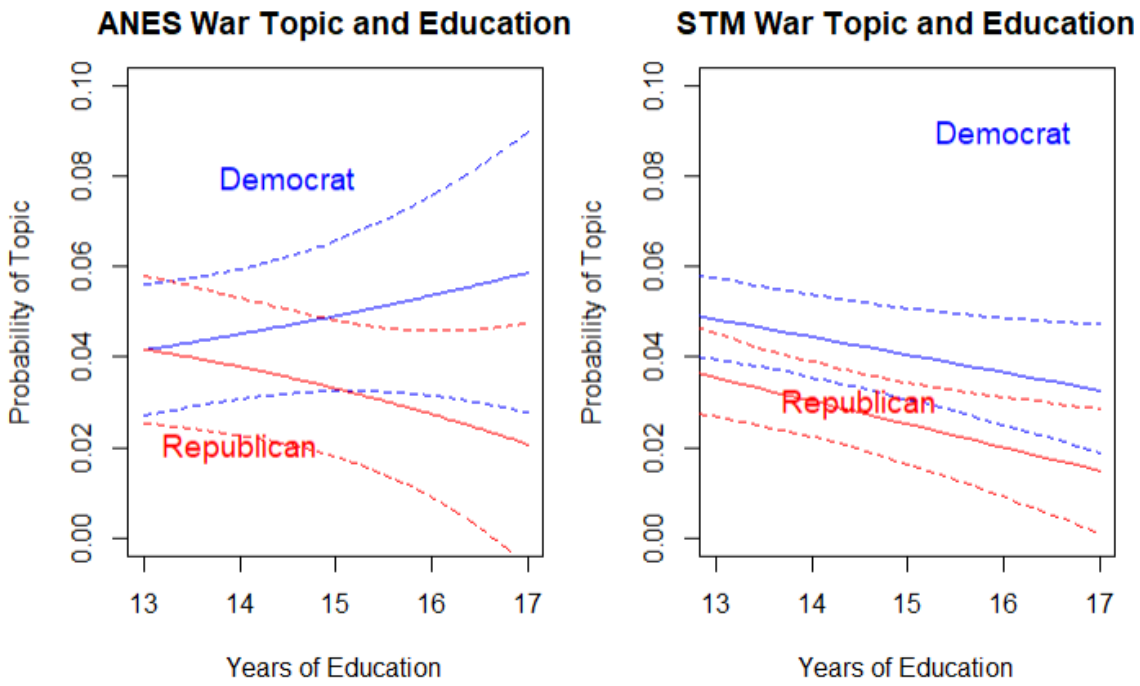


# ANES Most Important Problem: Figure 18




Original




Replication

# Replication Autopsy: What did I learn?

- Archived scripts relied on helper files from the developer's computer and old function names (so I had to do a bit of translating)
- Replication code appeared to be using a development STM version
- I chose models that were as close as possible to those chosen by the authors, but they were not exact
- I had to do a bit of digging and compatibility checking to match functions with archived versions of STM

# Replication Autopsy: What worked

- I was often able to replicate the substantive patterns from the original paper
- When using the modern version of STM, many of the functions were simple to update (just slightly changed names)

# Replication Autopsy: What did not work

The original data relied on developer helper scripts that were inaccessible/not included in the replication package or on the developer's Github

Similarly, the code relied on a development/early version of STM (which I also attempted to recover, but was unsuccessful)

Even when using the original seeds and models, exact model replication was not always possible, which led to different results

There was internal inconsistency in the metadata and vocab files, which I could not fix without adding new/random data to approximate (which is straying pretty far from replicating)

# Replication Autopsy: Why the Rand replication failed

- I attempted to replicate the workflow for the experiment, but was unable to run the model using the files in the replication package.
- The issue was a mismatch across files:
  - the document and metadata counts and a missing document feature IDs in the vocab file

# Extension

- More transparent model selection
  - Use new diagnostics to show more clearly why one topic solution was chosen over another
- Incorporate explicit stability checks
  - To show whether similar topics reappear across different seeds and candidate models
- Making preprocessing steps more explicit
  - STM now includes textProcessor() and prepDocuments() and
- Better replication packaging
  - I would include all helper scripts, software versions, and code for every figure (an in organized script) so the paper is easier to reproduce
- It may be interesting to make this process more automated, for example:
  - Create a function that both changes the seed and runs the candidate models itself? And then produces semantic coherence exclusivity plots that I can see and compare all at once.

# Conclusion

- Overall, I was able to partly replicate the core findings of the paper, but due to the missing replication files and inconsistent/improperly formatted files that were included for the Rand analysis on priming and intuition, I have not beeen able to reproduce a full replication of the paper.